

Cooper Daniel Trahan

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<https://coopertrahan.github.io/RedStegosSite/>

Education

University Of Memphis, Memphis, TN

2024 - Present

Bachelors, Computer and Electrical Engineering, Bachelors In Mathematical Sciences

- Currently enrolled in the University Of Memphis as a double major for Computer/Electrical Engineering and Mathematical sciences.
- Striving to seek out a Masters Degree in either one, and possibly a degree in Astro-Physics

Arlington High School - Diploma, Arlington TN

2020-2024

Computer Science Focus, - 3.5 GPA

- Distinction: One of only two students in school history to receive the Global Seal of Bi-Literacy in French.
- Technical Proficiency: Earned CIW certifications in JavaScript, HTML, and CSS while maintaining a 3.5 GPA.
- Leadership: Active member of Skills USA and AEI; attended NASA Space Camp (2023) to further interest in aerospace engineering.

Projects

Biomedical Systems Lab: Ear-Based Biopotential Sensing (OpenEarable ExG) | University of Memphis |

Present – December 2026

- **Role:** Taking over and advancing a research platform for OpenEarable ExG (an open-source hardware system for in-ear EEG, EOG, and EMG sensing).
- **System Integration:** Analyzing legacy schematics and PCB designs to integrate 7-channel biopotential sensing with the OpenEarable core (Nordic nRF52840 + AD7124-4 ADC).
- **Signal Processing:** Developing firmware and Python scripts to capture high-fidelity biopotential signals (EEG/EOG/EMG) via Bluetooth Low Energy (BLE) and USB Serial, focusing on noise reduction and artifact filtering.

Balloon CubeSat RCS Wheel System | University of Mississippi |

August - Present

- **Role:** Programmer of main logic behind RCS Wheel system for CubeSatProgram
- **Abstract:** A collaborative contribution to Mississippi State University's atmospheric research program. Assisting in design and data analysis of payloads launched via high-altitude weather balloons to study atmospheric conditions

3-DOF Robotic Arm with Hybrid Actuation | Independent / University Partnership (Michigan Univ.) |

May 2026 – August 2026

- **Design & Fabrication:** Engineered a 3-degree-of-freedom robotic arm using custom 3D-printed PLA components and precision-machined joints, achieving a **19-inch reach** and **2kg payload capacity**.
- **Control Systems:** Integrated a hybrid actuation system combining **stepper motors** for precise positioning and **servo motors** for dynamic movement, controlled via a custom microcontroller (e.g., Arduino/ESP32).
- **Kinematics:** Calculated forward and inverse kinematics to map joint angles to end-effector coordinates, ensuring smooth trajectory execution and payload stability.

Professional Experience

Undergraduate Research Assistant | University of Memphis / Memphis Biomedical Sensors & Systems Lab |

September 2026 – Present

- Lead development of **OpenEarable ExG**, integrating 7-channel **EEG/EOG/EMG** sensing using **Nordic nRF52840** and **AD7124-4** ADC hardware.

Technology Intern | Collierville Schools |

May 2025 – July 2026, May 2026 – July 2026

- Offered personalized tech support for students and teachers during summer school. Repaired and distributed 500+ apple devices across schools and pushed system wide updates and software

Certifications

CIW Javascript Specialist Certified (2023)
CIW HTML & CSS Specialist Certified (2023)
Global and State Seal of Bi-Literacy (French)

Skills

Python, HTML5, JavaScript, Git, Project Management, Arduino, Soldering, Circuit Design, PCB Design, Control Systems, Data Analysis, Rapid Prototyping, Signal Processing